

The `mstl.multistep` Champion Model

MSTL $\rightarrow$ ARIMA $\rightarrow$ Random-Forest Residual,
with a Location-Scale Variance Head

Model configuration: `config_arima012.bank`

June 5, 2026

Abstract

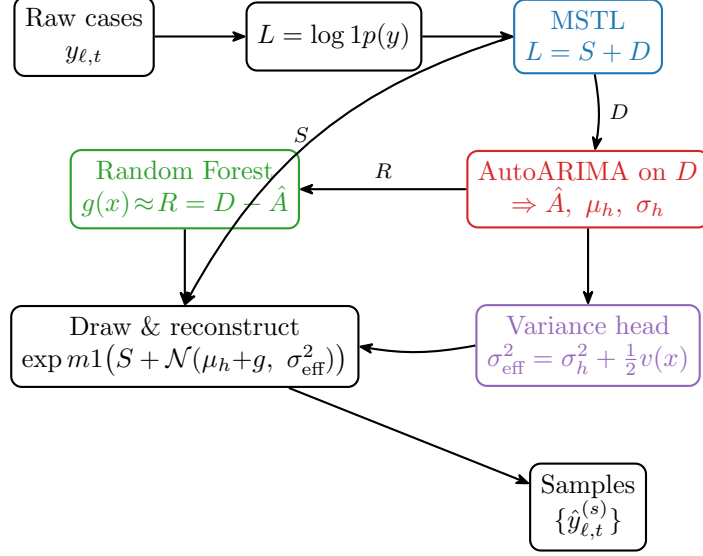
This document describes the champion forecasting model in the `mstl.multistep` project: a CHAP-compatible monthly disease-case forecaster. It is an *additive decomposition* model on the log scale. Multiple Seasonal-Trend decomposition (MSTL) removes the annual cycle; a **shared fixed-order ARIMA**(0, 1, 2) (the same order for every location) models the deseasonalised dynamics *and* supplies the predictive spread; a Random Forest corrects the ARIMA residual using lagged climate anomalies, an **expanded bank of indoor-residual-spraying (IRS) features** (incl. per-chemical decay channels), and lagged self-history; and a *location-scale variance head* widens the predictive interval by the Random Forest’s own uncertainty. On the frozen evaluation harness (406 locations, 3-month horizon) it reaches a log-CRPS of **0.3123** and CRPS of **81.88** — a $\sim 4\%$ log-CRPS improvement over the climate-only baseline.

1 Overview

The model forecasts monthly disease cases $y_{\ell,t}$ for each location ℓ . It works entirely in log $1p$ space, $L_{\ell,t} = \log(1 + y_{\ell,t})$, and writes that signal as a sum of three estimated components plus noise:

$$L_{\ell,t} = \underbrace{S_{\ell,t}}_{\text{seasonal (MSTL)}} + \underbrace{A_{\ell,t}}_{\text{AR / trend (ARIMA)}} + \underbrace{g(x_{\ell,t})}_{\text{covariate correction (RF)}} + \varepsilon_{\ell,t}. \quad (1)$$

Each component is fit by a different estimator chosen for what it is good at: MSTL for a stable repeating seasonal shape, ARIMA for autoregressive persistence *and* calibrated uncertainty, and a Random Forest for nonlinear, climate- and intervention-driven anomalies that the univariate ARIMA cannot see. The probabilistic forecast is Gaussian in log $1p$ space and is transformed back with $\exp m1$.



2 Data and notation

Input is a CHAP-style monthly panel: one row per (location, month) with the target `disease_cases`, climate covariates, an IRS allocation column, and an admin `district` id. There are 406 locations (sectors) over 2013–2026.

Symbol	Meaning
$y_{\ell,t}$	disease cases at location ℓ , month t
$L_{\ell,t} = \log 1p(y)$	log-transformed target (modelling space)
$S_{\ell,t}$	seasonal component (MSTL, period 12)
$D_{\ell,t} = L - S$	deseasonalised series (trend + remainder)
$\hat{A}_{\ell,t}$	ARIMA in-sample fit of D ; μ_h, σ_h its h -step forecast
$R_{\ell,t} = D - \hat{A}$	ARIMA residual (the RF target)
$g(\cdot)$	Random Forest mapping features x to a residual correction
$v(x)$	inter-tree variance of g at x (epistemic uncertainty)

3 Step 1 — MSTL seasonal decomposition

For each location independently, the log $1p$ series is decomposed by MSTL into a seasonal component (annual period $m = 12$) and a deseasonalised remainder:

$$L_{\ell,t} = S_{\ell,t} + D_{\ell,t}.$$

S captures the stable repeating within-year shape; $D = \text{trend} + \text{remainder}$ is what the rest of the pipeline forecasts. Into the forecast window the seasonal piece is extrapolated *seasonal-naive* (the last estimated annual cycle tiled forward). Series shorter than two full seasons are treated as all-trend ($S \equiv 0$).

4 Step 2 — Shared fixed-order ARIMA base (mean and spread)

A separate ARIMA is fit per location to the deseasonalised series D (with `season_length` = 1, since seasonality is already removed), but **all locations share the same fixed order**, **ARIMA(0, 1, 2)**

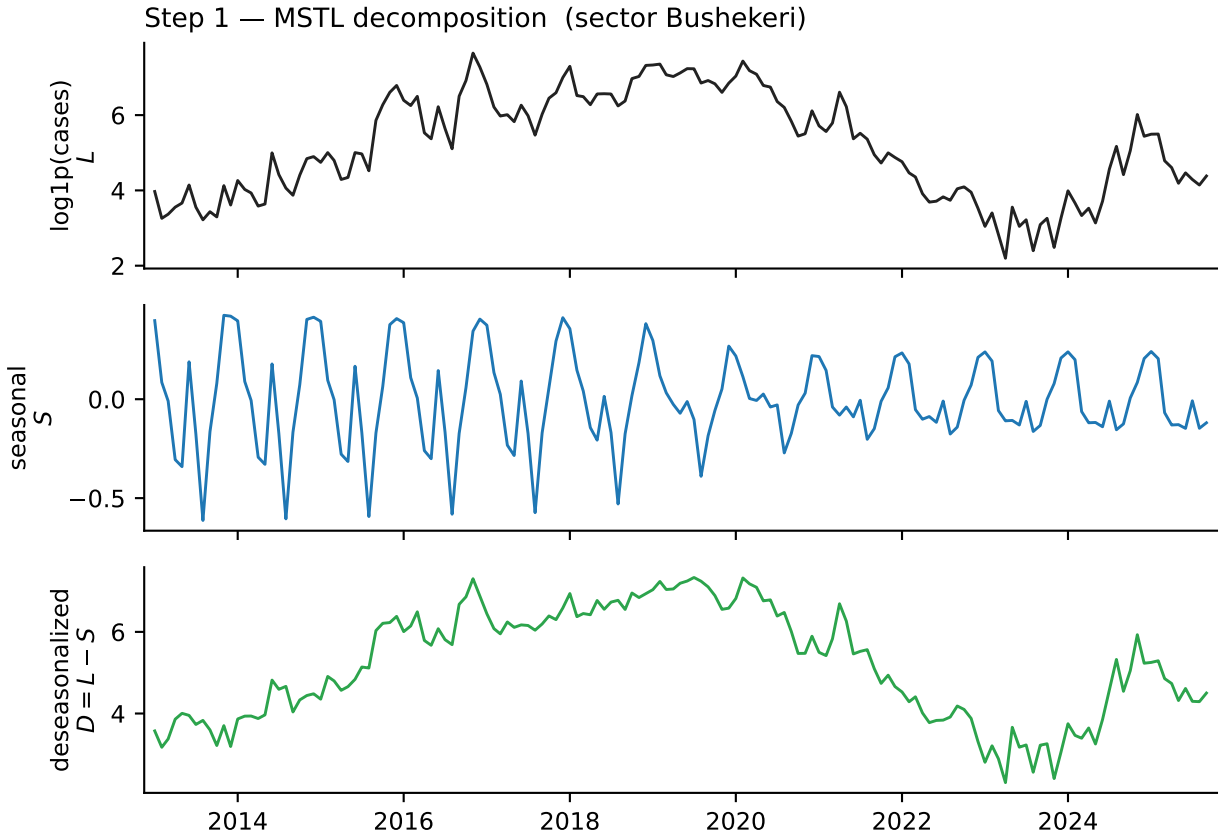


Figure 1: Step 1 in action for one sector. The $\log 1p$ case series L (top) is split by MSTL into a stable annual seasonal component S (middle) and the deseasonalised series $D = L - S$ (bottom) that the rest of the pipeline forecasts.

— only each location’s coefficients are estimated. This replaced per-location AutoARIMA order selection, which on short (157-month) noisy series is itself a source of noise (AutoARIMA overfits high-order/drift picks); a single robust order generalises better and was the largest single improvement found (-2% log-CRPS). See `arima_in_model.pdf` and `arima011_math.pdf` for the order analysis and the SES interpretation. ARIMA plays two roles:

1. **Autoregressive mean.** The h -step forecast mean $\mu_{\ell,h}$ carries persistence/trend in D . Its in-sample one-step fit $\hat{A}$ defines the residual $R = D - \hat{A}$ that the Random Forest then models.
2. **Predictive spread.** The forecast interval at level 68% is converted to a Gaussian standard deviation $\sigma_{\ell,h} = (\text{hi} - \text{lo})/(2z)$. This σ_h *grows with horizon* and is the model’s principal source of forecast uncertainty.

A degenerate series (no finite history) falls back to a constant-mean / sample-sd baseline.

5 Step 3 — Random-Forest residual correction

A single *pooled* Random Forest g is trained across all locations to predict the ARIMA residual $R_{\ell,t}$ from a feature vector $x_{\ell,t}$. It is a deterministic *point* correction — it nudges the mean, exploiting nonlinear structure ARIMA misses. The feature vector combines four blocks:

1. **Lagged climate anomalies.** The three climate covariates are first MSTL-deseasonalised (so the model sees *anomalies* relative to each one’s own seasonal cycle), then lagged with *per-covariate* windows reflecting the physical delay from weather to transmission:

covariate	lags (months)
<code>rainfall_era5</code>	1–6
<code>relative_humidity</code>	1–4
<code>mean_temperature</code>	1–2

Moisture-driven drivers (rain, humidity) get longer windows than temperature.

2. **Engineered IRS feature bank.** The raw `irs_allocated` column is a sparse 0–1 coverage fraction firing only in campaign months ($\sim 2.5\%$ of rows); its protective effect persists and is *known in advance*, so it is expanded into a bank of dense, contemporaneous (lag-0) signals and the forest selects what matters:

- `level`, since (months since last campaign, capped), `cumulative` (campaign-count stock);
- **per-chemical decay channels** — the geometric-decay protection split into `decay_{carbamate,pyrethro}` each non-zero only while that insecticide class is the active one (from `irs_insecticide_used`), so the RF learns each chemical’s effect magnitude (organophosphate/clothianidin strongest; pyrethroid near-dead, consistent with resistance);
- **decay basis** `decay2`, `decay8` (half-lives 2 and 8 months);
- **recency** flags `recent3/6/12` and `rounds12` (campaigns in the trailing year).

(The earlier 4-feature set `level/decay/since/cumulative` with a fixed 4-month decay was the prior champion; the bank improved on it.)

3. **Self-history (target lags).** The deseasonalised target D lagged 1–3 months. In the forecast window, lags that fall ahead of the origin are *bridged* with the ARIMA mean forecast μ_h (a non-recursive bridge), letting the RF pick up residual autocorrelation the linear ARIMA leaves behind.
4. **Location identity.** One-hot location dummies, so the pooled forest can retain per-series level/behaviour.

6 Step 4 — Location–scale variance head

The Random Forest correction is added as a point, so on its own it injects no uncertainty. The variance head restores that: the forest’s **inter-tree variance** $v(x) = \text{Var}_{\text{trees}} g(x)$ — how much the 200 trees disagree — is an estimate of the correction’s uncertainty, and it is folded into the predictive spread in quadrature:

$$\sigma_{\text{eff},\ell,h}^2 = \sigma_{\ell,h}^2 + \kappa v(x_{\ell,h}), \quad \kappa = 0.5. \quad (2)$$

This is the honest predictive variance “total = ARIMA forecast variance + variance of the mean correction.” It corrects a mild under-dispersion and improves *both* log-CRPS and CRPS — a rare non-trade-off in this project. (It is the productive form of an EM/coordinate-descent idea: a mean-backfitting EM was a documented negative because it reshuffled mean variance; moving the descent to the *spread* coordinate is what worked.)

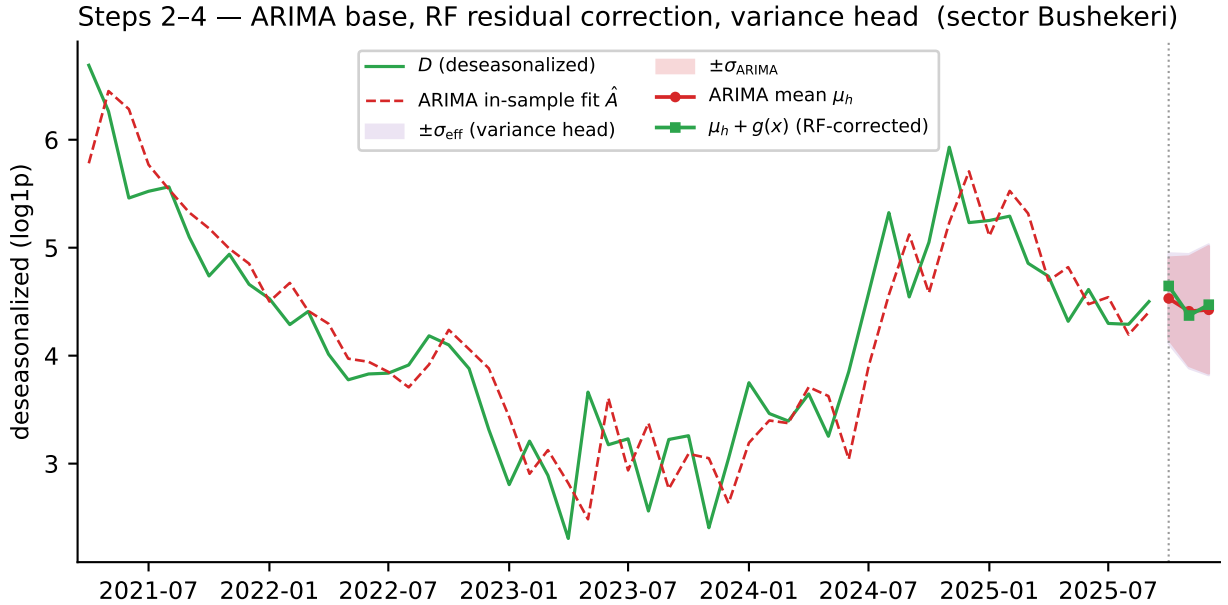


Figure 2: Steps 2–4 for the same sector, zoomed on the forecast origin (dotted line). The ARIMA fits the deseasonalised series D in-sample ($\hat{A}$, dashed) and forecasts the mean μ_h with spread σ_{ARIMA} ; the Random Forest shifts the mean by the correction $g(x)$ (green markers); the variance head widens the predictive spread from σ_{ARIMA} to σ_{eff} (purple band). (Figures 1–3 illustrate the shared pipeline; they were rendered on an immediately-prior champion config — the structure is identical.)

7 Step 5 — Reconstruction and sampling

For each location and horizon step h , the forecast distribution in $\log 1p$ space is Gaussian, centred on seasonal + ARIMA mean + RF correction, with the widened spread:

$$\hat{L}_{\ell,h}^{(s)} = S_{\ell,h} + \mu_{\ell,h} + g(x_{\ell,h}) + \eta^{(s)}, \quad \eta^{(s)} \sim \mathcal{N}(0, \sigma_{\text{eff},\ell,h}^2), \quad (3)$$

drawn $s = 1, \dots, 100$ times. Samples are mapped back to the count scale and clipped at zero:

$$\hat{y}_{\ell,h}^{(s)} = \max(0, \exp m1(\hat{L}_{\ell,h}^{(s)})).$$

The 100 samples per (location, month) are the model’s probabilistic forecast, scored by log-CRPS (primary) and CRPS.

Step 5 — reconstructed probabilistic forecast vs. actual

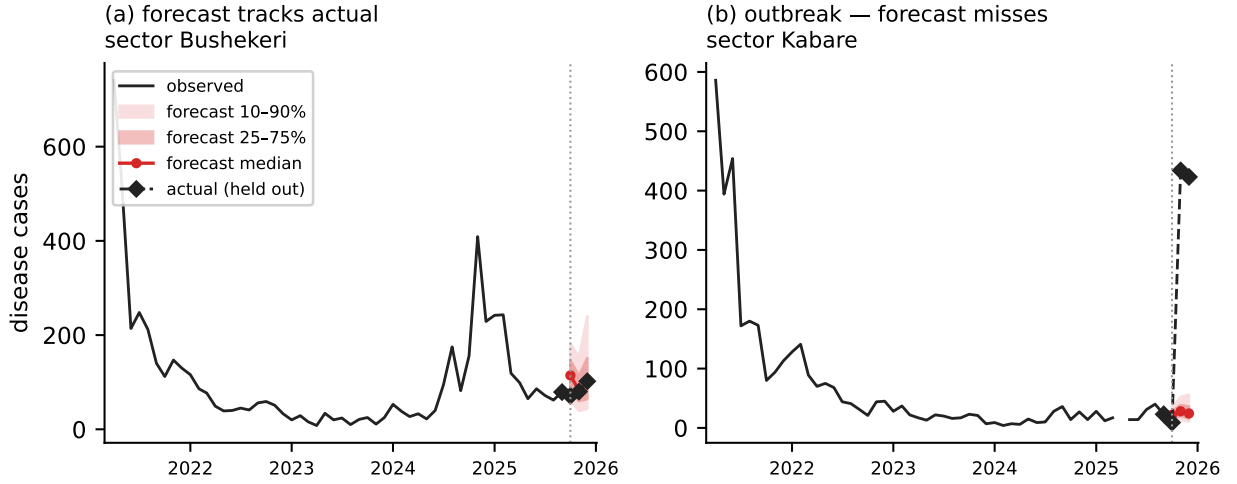


Figure 3: Step 5 on the count scale, for two sectors. **(a)** In a typical month the reconstructed forecast (median, with 10–90% and 25–75% bands) tracks the held-out actual. **(b)** For an explosive outbreak (actual $\approx 7\times$ the 90% band) the model under-predicts: it captures climate- and season-driven dynamics but not idiosyncratic outbreaks, which are the dominant remaining source of error and the reason the variance head (which widens the interval where the forest is unsure) helps but cannot fully cover the tail.

8 Champion configuration

Group	Setting	Value
Transform	<code>log_transform</code>	true (log $1p$)
	<code>season_length_monthly</code>	12
Base model	<code>prob_model</code>	<code>rf_residual</code>
	<code>arima_order</code>	[0, 1, 2] (shared, all locations)
	<code>arima_level</code>	68
Climate lags	<code>rainfall_era5</code>	1–6
	<code>relative_humidity</code>	1–4
	<code>mean_temperature</code>	1–2
	<code>deseasonalize_covariates</code>	all three
IRS	<code>irs_column</code>	<code>irs_allocated</code>
	<code>irs_chemical_column</code>	<code>irs_insecticide_used</code>
	<code>irs_features</code> (bank)	level, since, cumulative, chem.channels, decay2, decay8, recent3/6/12, rounds12
Self-history	<code>rf_target_lags</code>	3
Variance head	<code>residual_variance</code>	<code>tree</code>
	<code>residual_variance_scale</code> (κ)	0.5
Random Forest	<code>n_estimators</code>	200
	<code>max_depth</code>	none
	<code>min_samples_leaf</code>	3
	<code>max_features</code>	sqrt
Sampling	<code>n_samples</code>	100

9 Performance and design notes

On the frozen main harness — 406 locations, monthly, 3-month horizon, 12 backtest splits, stride 1 — the champion scores:

Model	log-CRPS ↓	CRPS ↓
Climate-only baseline (no IRS)	0.3253	83.6
+ IRS features, per-covariate lags, leaf= 3, target lags	0.3203	82.97
+ variance head (<code>config_tgtlag3_var</code>)	0.3200	82.88
+ shared ARIMA(0,1,2) order (<code>config_arima012</code>)	0.3135	81.77
+ IRS feature bank (champion , <code>config_arima012_bank</code>)	0.3123	81.88

What drives the model’s accuracy. The largest single lever was switching from per-location AutoARIMA *order selection* to one **shared fixed ARIMA**(0,1,2) order (−2% log-CRPS): order selection on short noisy series is itself noise. On top of that, the levers that helped all either *add information* (the IRS feature bank incl. per-chemical channels; lagged self-history) or *account for uncertainty honestly* (the variance head). A long list of alternatives — EM/backfitting on the mean, Fourier seasonal features, σ rescaling, quantile-GBM residuals, horizon-scaled variance, vegetation indices, spatial (same-district) neighbour features, chemical-*rate* decay, and reduced location encodings (district/cluster) — were tested and did not help. The recurring lesson: let each component own what it is best at (MSTL the season, ARIMA the persistence *and the spread*, the forest the nonlinear anomalies), prefer one robust shared configuration over per-series selection on short data, add information rather than re-shuffle it, and account for every component’s uncertainty honestly.